M2 Mid-Submission

1) Current approach and changes since M1:

Our project, MemoryTag AI, aims to develop an AI-powered system that automatically suggests relevant categories for images uploaded to Wikimedia Commons. The system must identify image content, generate suitable category suggestions, and rank them according to relevance and specificity.

We explored a computer-vision-based approach using a pretrained ResNet model for image classification. However, it had limited ability to describe diverse scenes, events, relationships, and abstract concepts. We therefore shifted toward a hybrid, Gemini-powered category suggestion pipeline that combines image understanding with Wikimedia Commons category retrieval.

The current approach uses Gemini to analyze uploaded images and generate structured information, including image descriptions, detected objects, possible categories, confidence scores, supporting evidence, specificity, and uncertainties. A ranking mechanism is intended to prioritize categories that accurately describe the image while avoiding overly generic or irrelevant suggestions. We improved the presentation of our model using a custom UI.

2) Experiments run so far and their results:

First, we tested Wikimedia Commons category retrieval using queries such as bicycle, bicycle in a city, motorcycle, and person riding a bicycle. The system successfully retrieved category-related results, including ten results for the bicycle query. This demonstrated that the Commons API could serve as the retrieval layer.

Second, we tested Gemini-based image analysis. After encountering a model availability error with gemini-2.5-flash, we switched to another model configuration. Image analysis and structured JSON generation subsequently worked, producing descriptions, objects, candidate categories, and supporting information.

The ranking stage did not yet produce a verified end-to-end result; sequential parent-category API requests appeared to cause delays.

Finally, we launched the Gradio interface successfully. However, subsequent integration tests displayed generic errors in the output components, indicating that the interface and backend still require debugging.

3) Progress on testing:

Testing has progressed from individual components toward full-system integration. Gemini image analysis and Commons retrieval have been demonstrated independently, while the interface has been launched. Nevertheless, a successful end-to-end operation has not yet been established.

Recent issues include undefined variables such as client, missing or incorrectly ordered function definitions, possible NumPy-to-PIL image conversion problems, and inconsistencies between the interface and backend.

Therefore, current results demonstrate partial functionality rather than a fully validated system. Quantitative evaluation using precision, recall, micro-F1, macro-F1, and mean average precision (mAP) remains necessary.

4) Plan for the remaining time:

The main priority is to cut down on the time taken to enlist all relevant categories. There are also some bugs in the current code which are leading to errors in certain cases, which is the immediate fix we need. We will then reduce unnecessary API calls, and handle empty results, timeouts, and uncertain predictions. Finally, we will compare suggested categories against manually selected reference categories and report appropriate evaluation metrics wherever sufficient labelled examples are available.

Our primary goal is to deliver a stable, demonstrable prototype with reproducible results and clearly documented limitations. This will allow us to establish a reliable baseline before pursuing more advanced ranking methods or large-scale evaluation.